Deep Learning (Master's degree in AI)

Practice 2 - RNNs (2025-2026)

P2 - ELECTRICITY LOAD FORECASTING

INSTRUCTIONS

Deadline

May 7th 23:59, 2026

Objectives

In this practice we will develop several deep learning models for time series forecasting, using real electricity consumption data from the state of New York.

Electricity demand forecasting is a relevant real-world problem, as it plays an important role in power system operation and energy market planning.

Dataset

The dataset used in this practice is based on real data from the New York Independent System Operator (NYISO).

The original data contain electricity load measurements for different zones in New York State. A preprocessing script has been used to download, clean and aggregate the data to hourly resolution, generating the CSV file provided for this practice. This script will be shared for reference.

The dataset includes data from 2021 to 2025, and contains:

- Hourly electricity load for the following zones: CAPITL, CENTRL, DUNWOD, GENESE, HUD VL, LONGIL, MHK VL, MILLWD, N.Y.C., NORTH, WEST.
- The corresponding total load.
- A Time Stamp column.

The problem to solve is defined as follows:

Given a sequence of past observations, predict the total load 3 hours ahead.

Part 1: Data pipeline and baselines

1. Data preprocessing

Prepare the data for training and evaluation:

- Split the dataset using a temporal split:
  - Training set: 2021–2023
  - Validation set: 2024
  - Test set: 2025
- Use as input features:
  - The individual zone loads.
  - The total load.
- Discard the Time Stamp column from the model inputs.
- Normalize the data:
  - Compute mean and standard deviation using only the training set.
  - Apply the same normalization to validation and test sets.
- Transform the data into input/output sequences using an appropriate sequence length.

The prediction target is the total load 3 hours ahead.

The evaluation metric is Mean Absolute Error (MAE). Models are trained on normalized data, but results must be reported as **denormalized MAE**.

2. Baseline models

Implement the following baselines:

- A last-value baseline, using the last observed value as prediction.
- A daily baseline, using the value corresponding to the same hour on the previous day.
- One neural baseline based on Dense layers, 1D convolutional layers, or a reasonable combination of both.

Part 2: Recurrent models and final analysis

1. Model development

Develop the following models:

- A model based on **SimpleRNN** layers.
- A model based on **LSTM** layers.
- A model based on **GRU** layers.

Each model must use only one type of recurrent layer, but it can be combined with other non-recurrent layers if appropriate.

You may explore relevant design choices such as the number of units, the number of layers, and regularization techniques (e.g., dropout).

You are not expected to perform an exhaustive hyperparameter search; focus on a limited number of meaningful experiments.

Try to use similar training configurations across models when possible (e.g., optimizer, batch size, learning rate), and justify any significant differences.

The configuration of each final model must be justified based on the experiments carried out.

2. Analysis and comparison of results

- Comment the results of each model developed.
- Summarize the final results in a **table or graph.**
- Make a reasoned comparison of all the results obtained, including both the recurrent models and the baseline models.
- Discuss the observed advantages and disadvantages of the different approaches, as well as any other relevant observations.

Practice groups

The practice will preferably be done in pairs (it can also be done individually). Both members of the group are responsible and should know everything that is delivered on their behalf.

No change of practice group will be made during the course. The group can be undone, but its members will continue to carry out the practice individually.

Submission

The exercises will be developed using Jupyter Notebooks.

Each notebook should include:

- The full names of the authors in the first cell.
- The complete ML process: data preparation, model definition, training and results.
- The results of execution.
- Commented code and explanatory text describing the process followed, the problems encountered, and the decisions taken.

If several notebooks are used, they must be submitted together in a ZIP file.

Submission will be done through the virtual campus platform of each university. The upload must respect the deadline.

- Universidade da Coruña: https://udconline.udc.gal/
- Universidade de Vigo: https://moovi.uvigo.gal/
- Universidade de Santiago de Compostela: https://cv.usc.es/

Evaluation criteria

Quality of the results

- MAE on the test set (denormalized).
- Comparison of models.
- Inclusion of relevant intermediate experiments.

Quality of the design

- Correct construction of the pipeline.
- Appropriate use of the different types of layers.
- Reasonable exploration of model configurations.

Quality of the explanations

- Clear description of the process.
- Justification of decisions.
- Correct interpretation of results.
- Comparison and summary of models in a final table or graph.